Cube-Sum Permutations: Unimodular Block Algebra and 57 Finite Shapes

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Abstract

Let $P(n)$ count permutations p of $[1,n]$ such that every $i+p(i)$ is a perfect cube. We prove $P(n)>0$ for every $n \geq 173$ from six explicit block-rearrangement families, 57 finite cube-word/output-order shapes, and elementary cubic suffix reflection. A general algebraic theorem proves that the block-order matrix has determinant 0 or 1. For each of the 57 shapes the determinant is 1, and a single Cramer-determinant formula yields strictly positive integral block lengths. The permutation then follows symbolically, without separately verifying 57 endpoint multiset equalities. Every such shape also admits a uniform infinite cubic dilation $N(h) = (N+\varepsilon)h^3 - \varepsilon$. A symbolic cubic-gap induction completes the existence tail for all $n \geq 365$. The 57 shape choices and positivity checks remain finite casework; this is an algebraic parameterization, not a case-free existence proof.

1 Two elementary lemmas

Lemma 1 (Boundary cancellation)**.** *Let $0 = b_0 < b_1 < \cdots < b_t = n$ and let $c_1, \ldots, c_t$ be positive integer cubes. Set $L_j = c_j - b_j$ and $R_j = c_j - b_{j-1}$. Assume $1 \leq L_j < R_j \leq n+1$. Then $p(i) = c_j - i$ for $b_{j-1} < i \leq b_j$ is a cube-sum permutation if and only if*

$$\boxed{\{1, R_1, \ldots, R_t\}_{\text{multi}} = \{n+1, L_1, \ldots, L_t\}_{\text{multi}}.}$$

Proof. Each input block maps bijectively onto the integer interval $[L_j, R_j)$. At an integer x, the jump in the total interval-coverage count is the multiplicity of x as a left endpoint minus its multiplicity as a right endpoint. The displayed identity makes that jump 1 at $x = 1$, -1 at $x = n+1$, and zero everywhere else. Thus coverage is exactly one at every $x \in [1,n]$. Reversing this argument proves necessity. □

Lemma 2 (Growing cubic suffix)**.** *Suppose $0 \leq m < n$, $m + n + 1 = c$ is a cube, and a cube-sum permutation on $[1,m]$ exists (the $m = 0$ permutation is empty). Keep the existing permutation on $[1,m]$ and set $p(i) = c - i$ for $m < i \leq n$. Then p is a cube-sum permutation on $[1,n]$.*

Proof. The new reflection is a bijection from $[m+1,n]$ to itself, and each new sum is c. □

2 Six algebraic block families and 57 exceptional identities

Lemma 3 (Reflected-block rearrangement)**.** *Choose positive input-block lengths $\ell_1, \ldots, \ell_t$ of sum n and an arbitrary output order π of these t blocks. Let A_j be the sum of input lengths before block j, and let B_j be the sum of output lengths before block j in order π. If all numbers*

$$C_j = 1 + A_j + B_j + \ell_j$$

are perfect cubes, then on input block j the reflection $p(i) = C_j - i$ gives a cube-sum permutation of $[1,n]$.

Proof. The input block is $[A_j + 1, A_j + \ell_j]$, which maps exactly to the prescribed output block $[B_j + 1, B_j + \ell_j]$. Those output blocks partition $[1,n]$ by construction. Each index sum is C_j. □

Family I (four blocks). Let $U < V$ be integer cubes and put

$$a = 2U - V - 1, \qquad b = 2V - 3U + 1.$$

If $a, b > 0$, use input lengths (a, b, a, b) and output order $(3, 4, 1, 2)$. Lemma 3 gives the cubic sums (U, V, U, V) and length $2(V - U)$. The pairs $(U, V) = (64, 125), (125, 216), (216, 343)$ yield lengths $122, 182, 254$ respectively.

Family II (five blocks). For cubes $X < Y < Z$, define

$$a = 2Y - Z - 1, \quad b = X + 2Z - 4Y + 1, \quad c = Y - X.$$

If $a, b, c > 0$, the lengths (a, b, c, a, c) in output order $(4, 2, 5, 1, 3)$ produce sums (Y, X, Z, Y, Z) and total length $2Y - X - 1$. With $(Y, Z) = (64, 125)$ and $X = 27, 8$, the results are $n = 100, 119$.

Family III (six blocks). For integer cubes X, W, Y, Z, put

$$\begin{aligned} a &= 2Y - Z - 1, & b &= X - 2a - 1, \\ c &= W + 2a - 2X + 1, & d &= Y + X - W - 2a - 1. \end{aligned}$$

If these are positive, choose lengths (a, b, c, d, a, d) and output order $(5, 2, 3, 6, 1, 4)$. The cubic sums are (Y, X, W, Z, Y, Z) and the length is $2Z - 2Y + X - W$. Taking $(Y, Z) = (64, 125)$, the choices $(X, W) = (8, 64), (27, 64), (8, 27)$ give $n = 66, 85, 103$, respectively.

Family IV (twelve blocks and three binary switches). Let X, T, Q be integers and put

$$\begin{aligned} u &= X - 7, & v &= T - 2X + 7, & w &= X + 27 - T, \\ z &= Q + 2T - 2X - 121, & h &= X + 152 - T - Q. \end{aligned}$$

Assume $u, v, z \geq 0$ and $w, h > 0$. Use input lengths

$$(3, u, v, w, 3, w, z, h, 3, 27, 3, h)$$

and output block order

$$(11, 2, 3, 6, 9, 4, 7, 12, 5, 10, 1, 8).$$

The reflected-block lemma computes the twelve nominal index-image sums as

$$(125, X, T, 64, 125, 64, Q, 216, 125, 216, 125, 216).$$

When the second, third, or seventh block is nonempty, its nominal sum X, T, or Q, respectively, must be a cube. Any zero-length block is omitted from both orders and imposes no cube condition. The total length is

$$n = 276 + X - T - Q.$$

Take $X \in \{7, 8\}$, choose $T \in \{2X - 7, 27\}$, and independently choose $Q \in \{121 - 2T + 2X, 125\}$. All eight resulting length choices satisfy the positivity and cube conditions. The 2^3 distinct lengths are

$$\boxed{131, 132, 150, 151, 155, 156, 174, 175}.$$

This is one symbolic rearrangement construction with three optional empty blocks, not eight separate finite endpoint certificates.

Family V (15 blocks, two constrained switches). Take $\alpha, \beta \in \{0, 1\}$ with $\alpha \leq \beta$ and $t \in \{0, 7 + 4\beta + 2\alpha\}$. Define the ordered input block lengths

$$\begin{gathered} \ell_1 = 2\beta, \quad \ell_2 = 7 - \alpha - 4\beta - \alpha\beta, \quad \ell_3 = \alpha, \quad \ell_4 = \alpha\beta, \\ \ell_5 = 13 + 4\beta, \quad \ell_6 = \alpha, \quad \ell_7 = 23 - 4\beta - \alpha, \quad \ell_8 = 13 + 4\beta, \\ \ell_9 = 48 - \alpha - 45\beta + \alpha\beta, \quad \ell_{10} = 2\beta, \quad \ell_{11} = \beta(39 - \alpha), \\ \ell_{12} = t, \quad \ell_{13} = 30 + \alpha - t, \quad \ell_{14} = 13 + 4\beta, \quad \ell_{15} = 91 + \alpha - t. \end{gathered}$$

The output block order is

$$(10, 4, 6, 2, 8, 3, 11, 1, 9, 14, 7, 12, 15, 5, 13).$$

All lengths are nonnegative, with empty blocks omitted. Because $\alpha\beta = \alpha$, the nominal reflected sums are

$$\begin{gathered} 21 + 43\beta,\ 8,\ 28 - \alpha,\ 8,\ 216, \\ 21 + 2\alpha + 4\beta,\ 125,\ 64,\ 125,\ 105 - 41\beta, \\ 125,\ 209 - 2\alpha - 4\beta + t,\ 343,\ 216,\ 343. \end{gathered}$$

Every *nonempty* block has a cubic sum: blocks $1, 10, 11$ are used only if $\beta = 1$; blocks $3, 4, 6$ are used only if $\alpha = \beta = 1$; and block 12 is empty if $t = 0$, otherwise its sum is 216. The total length is

$$\boxed{n = 238 + \alpha + 2\beta - t},$$

so the three allowed pairs $(\alpha, \beta) = (0,0), (0,1), (1,1)$ with their two possible t values give

$$\boxed{228, 229, 231, 238, 240, 241}.$$

This is a symbolic cubic-sum construction; it does not require verifying six separate endpoint lists.

Family VI (ten blocks, nested binary degeneration). Let $\alpha, \beta \in \{0, 1\}$ with $\beta \le \alpha$. Use input block lengths

$$(7,\ 12\alpha,\ 159 - 24\alpha - 99\beta,\ 14\beta,\ 28\beta,\ 63\beta,$$
$$10 + 24\alpha - 6\beta,\ 8\beta,\ 159 - 24\alpha - 107\beta,\ 10 + 24\alpha + 93\beta)$$

and output order $(1, 2, 8, 5, 6, 9, 4, 10, 3, 7)$. Direct substitution into the reflected-block lemma gives the ten nominal sums

$$8,\ 8 + 19\alpha,\ 343,\ 167 - 12\alpha + 61\beta,\ 167 - 12\alpha - 30\beta,$$
$$167 - 12\alpha + 61\beta,\ 512,\ 177 + 12\alpha + 27\beta,\ 343,\ 512.$$

Block 2 is nonempty only when $\alpha = 1$ and has cubic sum 27; blocks $4, 5, 6, 8$ are nonempty only when $\beta = \alpha = 1$ and have sums $216, 125, 216, 216$. Other nonempty blocks have constant cube sums; all empty blocks are omitted. Since

$$n = 345 + 12\alpha - 6\beta,$$

the three admissible choices give the lengths $345, 357, 351$, using one parametric construction rather than three independent endpoint lists.

2.1 Universal unimodular determinant construction

The remaining 57 integer witnesses admit a common algebraic formula, without separately testing their endpoint cancellation identities. For an output-block order π on t labels, write $\rho(i) = \pi^{-1}(i)$ and set

$$M_{ji} = [i < j] + [\rho(i) < \rho(j)] + [i = j], \quad 1 \le i, j \le t.$$

Here brackets denote indicator functions.

Lemma 4 (Unimodular block-order matrices)**.** *For every finite output-block order π,* $\det M \in \{0, 1\}$.

Proof. With $\mathbf{1}$ denoting the all-ones vector, write $M = \mathbf{1}\mathbf{1}^T + K$. The matrix K is skew-symmetric, and for $i < j$ its (i, j) entry is -1 exactly when $\rho(i) < \rho(j)$, and 0 otherwise. It is therefore an oriented chord-intersection matrix: place first chord endpoints in input order around one arc of a circle, and second endpoints in output order around the complementary arc.

Recall that an even-order chord-intersection matrix has Pfaffian 0 or ± 1. Indeed, thickening the chord diagram into an oriented ribbon surface, the chord loops form an integral homology basis. Their intersection form is integrally congruent to a direct sum of unimodular symplectic pairs $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and zero blocks from the boundary radical. Its full Pfaffian is thus 0 or ± 1.

If t is even, $\operatorname{adj}(K)$ is skew-symmetric; the rank-one determinant identity gives

$$\det M = \det K + \mathbf{1}^T \operatorname{adj}(K)\mathbf{1} = \operatorname{Pf}(K)^2 \in \{0, 1\}.$$

If t is odd, $\det K = 0$. Adjoin a new chord crossing all original chords, whose skew matrix is

$$H = \begin{pmatrix} 0 & \mathbf{1}^T \\ -\mathbf{1} & K \end{pmatrix}.$$

The same rank-one and bordered determinant identities give

$$\det M = \mathbf{1}^T \operatorname{adj}(K)\mathbf{1} = \det H = \operatorname{Pf}(H)^2 \in \{0, 1\}.$$

□

Theorem 1 (Symbolic determinant families)**.** *Fix π with $\det M = 1$. For arbitrary integer cube roots $k_1, \ldots, k_t \geq 2$, let $d = (k_j^3 - 1)_{j=1}^t$. Define*

$$\ell_j = \det M^{(j)}(d),$$

where $M^{(j)}(d)$ is obtained from M by replacing column j with d. If every $\ell_j > 0$, then with $N = \sum_j \ell_j$ there is a permutation of $[1, N]$ with cube sums k_j^3 on successive input blocks of these lengths.

Proof. By the preceding lemma and Cramer's rule, $M\ell = d$ holds identically over the integers. If A_j, B_j are the input and output block prefix lengths, this is precisely

$$k_j^3 = 1 + A_j + B_j + \ell_j.$$

Reflection $p(i) = k_j^3 - i$ maps the jth input block bijectively to the jth block in output order π. The output blocks partition $[1, N]$. □

The 57 cubic-root words and block orders in Appendix B are therefore a collection of *symbolic* families, rather than 57 independently postulated endpoint identities. For each printed root word, integer determinant evaluation gives $\det M = 1$, $\ell_j \geq 1$, and the specified length N (the smallest ℓ_j across the full table equals 1). The separate exact `verify_unimodular_factory.py` checks these finite positivity conditions. The 57 parameter choices remain finite data: this is not a case-free proof of their existence.

Theorem 2 (Uniform cubic dilation)**.** *Under Theorem 1, put $v = M^{-1}\mathbf{1}$ and $\varepsilon = t \bmod 2$. For every integer $h \geq 1$,*

$$\ell_j(h) = \ell_j + (h^3 - 1)(\ell_j + v_j) > 0$$

defines another cube-sum permutation, with root word $(hk_1, \ldots, hk_t)$, of length

$$\boxed{N(h) = (N + \varepsilon)h^3 - \varepsilon.}$$

Proof. Pfaffian minor expansions in the preceding chord-diagram proof imply $v_j \in \{-1, 0, 1\}$: the Cramer minors are signed Pfaffians of chord diagrams with a marked chord deleted, divided by a nonzero Pfaffian of absolute value 1. Moreover $\sum_j v_j = \varepsilon$. Indeed, $Mv = \mathbf{1}$ and skew symmetry give $S^2 = S$ for $S = \mathbf{1}^T v$. If t is even, K is invertible and $v = K^{-1}\mathbf{1}$ yields $S = 0$. If t is odd, invertibility of M and singularity of K exclude $S = 0$: a kernel vector of K has nonzero pairing with $\mathbf{1}$. Thus $S = 1$.

Since $\ell_j \geq 1$ and $v_j \geq -1$, all displayed scaled lengths are positive. Finally

$$M\ell(h) = h^3 M\ell + (h^3 - 1)\mathbf{1} = h^3 (k_j^3)_{j=1}^t - \mathbf{1}.$$

The determinant construction applies again. Summing the lengths gives the formula for $N(h)$. □

Lemma 5 (Affine-word rigidity)**.** *Fix a finite list of integer cubes $C_1, \ldots, C_t$. Suppose all breakpoints $b_j(n)$ are affine functions of n with $b_0(n) = 0$ and $b_t(n) = n$. The resulting fixed-cube reflection word can be a valid permutation for at most two distinct values of n.*

Proof. Every legal permutation satisfies the sum identity

$$\sum_{j=1}^{t} C_j \big(b_j(n) - b_{j-1}(n)\big) = \sum_{i=1}^{n} (i + p(i)) = n(n+1).$$

Its left side is affine in n; its right side is a monic quadratic. The equality cannot hold for three distinct values. □

The obstruction concerns *fixed* cube sums and affine endpoints only, not varying cube sums or composition of reflection templates.

In total these *six symbolic identities* give twenty-five base lengths

$$66, 85, 100, 103, 119, 122, 131, 132, 150, 151, 155, 156, 174, 175, 182, 228, 229, 231, 238, 240, 241, 254, 345, 351, 357$$

without twenty-five separate endpoint-table checks.

The remaining 60 rows. The appendix records 57 exceptional rows of the form $n\ b_1 : k_1\ \cdots\ b_t : k_t$. The corresponding cubical sums are $c_j = k_j^3$ and the first boundary is $b_0 = 0$. For every row, the boundaries increase, $1 \le L_j < R_j \le n+1$, and the multiset equality in Lemma 1 holds exactly. These are 795 simple integer-interval identities. Exactly 10 listed rows have $n < 173$ and exactly 47 have $173 \le n \le 364$. Together with the twenty-five algebraic-template lengths, there are 22 lower and 60 upper seeds.

The following two lists give every omitted base length without further matching data. For any tabulated predecessor m under cube c, define $n = c - m - 1$ and apply Lemma 2. The lists are ordered by cube, so every predecessor is either a seed or was obtained in an earlier row.

First: 26 reflected lengths below 173.

```
cube     already constructed predecessors m
  8      0
 27      0,7
 64      0,7,19,26
125      0,7,19,26,37,44,56
216      44,56,63,66,68,80,82,85,87,
         100,103,105
```

Together with the 10 exceptional lower seeds and the twelve algebraic-template lower seeds, these supply 48 distinct lengths below 173.

Second: 49 reflected lengths in $[173, 255] \cup [339, 364]$.

```
cube     previously constructed predecessors m
216      0,7,19,26,37
343      0,87,98,100,103,105,110,112,
         115,116,117,118,119,120,121,122,
         123,124,125,126,128,129,130,131,
         132,133,135,147,149,150,151,152,
         155,156,159
512      147,149,150,151,152,155,156,
         159,171
```

The 47 upper exceptional rows, the thirteen upper algebraic-template lengths and these 49 reflections are exactly the 109 integers in $[173, 255] \cup [339, 364]$. There is no circular dependency: each predecessor m is strictly smaller than its constructed length n.

For example, $211 = 343 - 131 - 1$, $340 = 512 - 171 - 1$, and $342 = 343 - 0 - 1$. The first example replaces a previously necessary-looking 53-segment independent row by one suffix reflection.

Lemma 6 (Complete finite interval)**.** *For every integer $173 \le n \le 364$, a cube-sum permutation of $[1, n]$ exists.*

Proof. The explicit arithmetic rows and the two predecessor tables establish the claim for $[173, 255] \cup [339, 364]$. For each of the 83 remaining lengths $256 \le n \le 338$, take $m = 511 - n \in [173, 255]$. Then $m < n$ and $m + n + 1 = 512 = 8^3$, so Lemma 2 applies. □

2.2 The interval-overlap avalanche and its finite obstruction

Lemma 7 (Exact cubic reflection of a good interval)**.** *Suppose every integer $m \in [a, b]$, $0 \le a \le b$, has a legal cube-sum permutation. For a cube c, every integer n in the nonempty interval*

$$\left[\max\{1, \lceil c/2 \rceil, c - b - 1\},\ c - a - 1\right]$$

also has a legal cube-sum permutation.

Proof. Take $m = c - n - 1$. By the displayed bounds, $a \le m \le b$ and $0 \le m < n$. Keep the old permutation on $[1, m]$ and set $p(i) = c - i$ on $[m + 1, n]$. The suffix is mapped bijectively to itself, with constant sum c. □

Lemma 8 (Interval-overlap avalanche)**.** *Suppose every length in $[a, b]$ is legal, where $1 \le a \le b$. If a cube c satisfies*

$$\boxed{a + b + 2 \le c \le 2b + 2},$$

then every length in the larger interval $[a, c - a - 1]$ is legal.

Proof. By Lemma 7, the new interval of lengths includes $b+1$ because $a \le c-b-2 \le b$ and $c \le 2(b+1)$. It continues through $c-a-1 \ge b+1$; it therefore meets the already established interval immediately, with no gap. □

Theorem 3 (Infinite avalanche from the finite base)**.** *If all lengths* $173 \le n \le 364$ *are legal, then every length* $n \ge 173$ *is legal, without any additional finite certificate.*

Proof. For $a = 173$, $b = 364$, the cube $9^3 = 729$ lies in the avalanche window $[539, 730]$. Consequently $[173, 555]$ is legal. Assume $[173, k^3 - 174]$ is legal for $k \ge 9$. The cube $(k+1)^3$ obeys

$$k^3 + 1 \le (k+1)^3 \le 2k^3 - 346,$$

since the last inequality is equivalent to

$$f(k) = k^3 - 3k^2 - 3k - 347 \ge 0,$$

with $f(9) = 112$ and $f(k+1) - f(k) = 3k^2 - 3k - 5 > 0$ for $k \ge 9$. Lemma 8 extends the good interval through $(k+1)^3 - 174$. Induction on k proves the assertion. □

Why overlap does not by itself remove the finite base. At $n = 65$, the domains $[1, 63]$ of the 64-reflection and $[60, 65]$ of the 125-reflection overlap and jointly cover $[1, 65]$. Yet

$$S = \{3, 40, 59, 64\}, \qquad N(S) = \{5, 24, 61\}$$

under the available cube sums $8, 27, 64, 125$, so Hall's inequality fails. Likewise no smaller legal length can produce $n = 173$ by a single suffix reflection: the only possible predecessors are $m = 42$ and $m = 169$, via cubes $216, 343$. For $m = 42$, the six-row set

$$[1, 2] \cup [20, 21] \cup [38, 39]$$

has just the four cube-sum neighbors $[6, 7] \cup [25, 26]$. For $m = 169$, the fourteen-row set

$$[8, 9] \cup [38, 39] \cup [45, 46] \cup [99, 100]$$
$$\cup [129, 130] \cup [136, 137] \cup [160, 161]$$

has only the twelve neighbors

$$[18, 19] \cup [25, 26] \cup [55, 56]$$
$$\cup [79, 80] \cup [86, 87] \cup [116, 117].$$

These are direct Hall obstructions. There is also no cube in the interval $[513, 678]$, the necessary avalanche window for extending $[173, 338]$ through 339. Thus the avalanche lemma completely removes the infinite tail's finite checks, but *does not derive the original* 173–364 *base interval without the separately constructed finite witnesses.* Such a claim would require an additional local augmentation or gluing theorem.

3 Existence for every length $n \ge 173$

Theorem 4. *For every integer* $n \ge 173$*, there is a permutation* p *of* $[1, n]$ *for which every* $i + p(i)$ *is a perfect cube.*

Proof. The finite interval $[173, 364]$ is provided by Lemma 6, from the six explicit block families and 57 algebraic cubic-order shapes. The interval-overlap avalanche Theorem 3 then propagates it to every integer $n \ge 173$. No further finite construction is required. □

Exact logical status. The infinite propagation is purely symbolic, including the full cube-gap inequality. Each of the 57 finite cubic-order shapes is given by an integer Cramer-determinant formula with a positive block-length vector, not by an appeal to a matching solver. The numerical root words, output orders, and positivity checks for these 57 shapes are still indispensable *finite case data.* The exact audit program `verify_unimodular_factory.py` is a reproducibility aid, not a hidden premise of the interval avalanche theorem. The present argument does *not* establish a case-free existence proof of the original base interval.

A Redundant 57-row endpoint witnesses

Every token $b : k$ means $p(i) = k^3 - i$ up to endpoint b. For each row, verify the same endpoint balance of Lemma 1.

82: 2:4 5:2 21:3 42:4 61:5 63:4 82:5

116: 2:4 3:3 4:2 8:3 9:5 10:3 11:5 15:3 16:5 17:3 18:5 22:3 23:5 24:3 61:5 63:4 100:5 101:6 102:5 106:6 107:5 108:6 109:5 113:6 114:5 115:6 116:5

118: 1:3 2:2 3:3 4:2 5:3 6:2 7:5 8:3 9:5 10:3 11:5 12:3 14:5 15:3 16:5 17:3 18:5 19:3 21:5 22:3 23:5 24:3 25:5 26:3 98:5 99:6 100:5 101:6 102:5 103:6 105:5 106:6 107:5 108:6 109:5 110:6 112:5 113:6 114:5 115:6 116:5 117:6 118:5

120: 2:4 3:3 4:2 9:5 10:3 16:5 17:3 23:5 24:3 61:5 63:4 100:5 101:6 107:5 108:6 114:5 115:6 120:5

121: 2:4 3:3 9:5 10:3 16:5 17:3 23:5 24:3 61:5 63:4 100:5 101:6 107:5 108:6 114:5 115:6 121:5

123: 2:4 31:5 32:4 61:5 63:4 92:5 93:6 122:5 123:6

125: 3:5 4:2 29:5 30:4 33:5 34:4 90:5 91:6 94:5 95:6 120:5 121:6 124:5 125:6

126: 2:4 5:2 28:5 35:4 61:5 63:4 89:5 96:6 119:5 126:6

127: 1:5 4:2 5:3 6:2 11:5 12:3 18:5 19:3 25:5 26:3 27:5 30:4 31:5 36:4 88:5 91:6 92:5 97:6 102:5 103:6 109:5 110:6 116:5 117:6 118:5 121:6 122:5 127:6

129: 3:5 4:2 8:3 11:5 15:3 18:5 22:3 25:5 30:4 33:5 38:4 86:5 91:6 94:5 99:6 102:5 106:6 109:5 113:6 116:5 121:6 124:5 129:6

173: 7:2 12:3 14:5 19:3 21:5 42:4 103:6 105:5 110:6 112:5 173:6

176: 7:2 12:3 14:5 17:4 19:3 24:5 42:4 103:6 105:5 110:6 115:5 166:6 169:7 176:6

177: 3:5 4:2 8:3 11:5 15:3 18:5 22:3 25:5 30:4 33:5 38:4 91:6 94:5 99:6 102:5 106:6 109:5 113:6 116:5 121:6 124:5 177:6

179: 3:5 11:3 15:4 23:3 30:4 33:5 40:4 48:6 52:4 91:6 94:5 121:6 124:5 163:6 167:7 175:6 179:7

180: 5:3 7:5 12:3 14:5 19:3 21:5 26:3 28:5 35:4 96:6 98:5 103:6 105:5 110:6 112:5 117:6 119:5 180:6

181: 3:5 4:2 29:5 30:4 33:5 34:4 91:6 94:5 95:6 120:5 121:6 124:5 181:6

184: 7:2 31:4 32:6 56:4 159:6 183:7 184:6

185: 21:5 26:3 30:4 51:6 58:4 91:6 112:5 178:6 185:7

188: 2:5 5:4 7:2 30:4 32:6 61:4 91:6 93:5 154:6 157:7 159:6 188:7

194: 7:2 12:3 14:5 19:3 21:5 103:6 105:5 110:6 112:5 194:6

197: 8:5 18:3 55:6 63:4 91:6 99:5 182:6 190:7 197:6

198: 2:4 5:2 28:5 35:4 53:6 61:5 63:4 71:5 96:6 119:5 144:6 162:7 180:6 198:7

199: 2:4 5:2 16:4 22:6 41:4 47:6 58:4 61:5 63:4 66:5 149:6 168:7 174:6 193:7 199:6

200: 26:3 37:4 51:6 73:5 142:6 164:7 178:6 200:7

201: 1:3 12:4 25:3 37:4 50:6 62:5 63:4 141:6 164:7 177:6 190:7 201:6

202: 13:4 25:6 38:4 50:6 63:4 152:6 165:7 177:6 190:7 202:6

203: 12:4 25:6 26:3 37:4 51:6 62:4 152:6 164:7 178:6 189:7 203:6

204: 11:4 15:6 26:3 36:4 51:6 61:5 142:6 163:7 178:6 188:7 204:6

205: 2:4 5:2 21:3 42:4 46:6 61:5 63:4 78:5 137:6 169:7 173:6 205:7

206: 7:2 18:3 19:4 20:6 44:4 45:6 55:5 56:4 79:5 136:6 146:7 147:6 171:7 172:6 206:7

233: 7:2 9:6 18:4 43:5 45:6 70:5 81:6 106:5 109:6 134:7 136:6 170:7 172:6 197:7 208:6 233:7

234: 16:6 36:5 52:6 72:5 88:6 108:5 127:7 143:6 163:7 179:6 199:7 215:6 234:7

235: 7:2 9:6 17:3 43:5 45:6 79:5 81:6 107:5 134:7 136:6 170:7 172:6 206:7 208:6 235:7

236: 7:2 18:6 43:5 45:6 54:4 79:5 81:6 106:5 134:7 145:6 170:7 172:6 206:7 208:6 236:7

243: 25:4 61:5 63:4 86:6 99:5 190:7 213:6 243:7

245: 27:4 61:5 88:6 97:5 188:7 215:6 245:7

246: 2:6 24:3 28:6 36:5 38:4 60:5 61:6 63:4 64:6 96:5 127:7 129:6 151:7 155:6 187:7 188:6 190:7 191:6 246:7

247: 2:6 3:2 4:3 5:2 22:6 23:3 29:6 36:5 38:4 41:5 58:4 59:5 61:6 63:4 65:6 95:5 127:7 129:6 132:7 149:6 150:7 156:6 186:7 188:6 190:7 192:6 247:7

248: 2:6 5:2 30:6 36:5 38:4 41:5 58:4 61:6 63:4 66:6 94:5 127:7 129:6 132:7 157:6 185:7 188:6 190:7 193:6 248:7

249: 1:3 6:2 31:6 42:5 57:4 62:6 63:4 67:6 93:5 133:7 158:6 184:7 189:6 190:7 194:6 249:7

250: 7:2 32:6 43:5 56:4 68:6 92:5 134:7 159:6 183:7 195:6 250:7

251: 7:2 19:3 33:6 55:5 69:6 91:5 146:7 160:6 182:7 196:6 251:7

252: 7:2 9:6 18:4 34:6 43:5 45:6 54:5 81:6 90:5 134:7 136:6 145:7 161:6 170:7 172:6 181:7 208:6 252:7

253: 2:4 24:3 25:6 27:4 28:6 35:4 60:5 64:6 86:5 88:6 89:5 151:7 152:6 154:7 155:6 187:7 191:6 213:7 215:6 253:7

339: 6:3 20:6 23:4 26:3 172:6 302:8 316:7 322:8 339:7

341: 1:3 21:7 62:6 63:4 189:6 190:7 317:8 341:7

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353: 6:3 7:2 20:6 25:3 43:5 57:4 116:6 134:7 158:6 285:8 316:7 353:8

354: 5:7 21:4 22:3 58:4 157:6 169:8 173:7 174:6 338:8 342:7 354:8

358: 6:3 7:2 43:7 62:4 85:5 153:6 176:8 189:6 212:7 358:8

363: 4:7 6:6 21:4 42:6 67:5 133:6 148:7 169:8 173:7 300:8 342:7 363:8

B The 57 algebraic cubic-order shapes

For each target n, the line gives a cube-root word $k_1, \ldots, k_t$ and an output-block order π. Its block lengths are not assumed or listed: they are the integer determinants in Theorem 1. The target length is a checksum for their sum. All listed lengths are strictly positive.

82: roots 4,2,3,4,5,4,5; order 6,2,3,4,7,1,5

116: roots 4,3,2,3,5,3,5,3,5,3,5,3,5,4,5,6,5,6,5,6,5,6,5,6,5; order 16,14,3,12,27,10,25,8,23,6,21,4,19,2,17,1,15,26,13,24,11,22,9,20,7,18,5

118: roots 3,2,3,2,3,2,5,3,5,3,5,3,5,3,5,3,5,3,5,3,5,3,5,6,5,6,5,6,5,6,5,6,5,6,5,6,5,6,5; order 24,6,22,4,20,2,43,18,41,16,39,14,37,12,35,10,33,8,31,5,29,3,27,1,25,42,23,40,21,38,19,36,17,34,15,32,13,30,11,28,9,26,7

120: roots 4,3,2,5,3,5,3,5,3,5,4,5,6,5,6,5,6,5,6,5; order 11,9,3,18,7,16,5,14,2,12,1,10,17,8,15,6,13,4

121: roots 4,3,5,3,5,3,5,3,5,4,5,6,5,6,5,6,5,6,5; order 10,8,17,6,15,4,13,2,11,1,9,16,7,14,5,12,3

123: roots 4,5,4,5,4,5,6,5,6; order 5,8,3,6,1,4,9,2,7

125: roots 5,2,5,4,5,4,5,6,5,6,5,6,5,6; order 13,2,11,6,9,4,7,14,5,12,3,10,1,8

126: roots 4,2,5,4,5,4,5,6,5,6; order 6,2,9,4,7,1,5,10,3,8

127: roots 5,2,3,2,5,3,5,3,5,3,5,4,5,4,5,6,5,6,5,6,5,6,5,6,5,6,5,6; order 10,4,27,2,25,8,23,6,21,3,19,14,17,12,15,28,13,26,11,24,9,22,7,20,5,18,1,16

129: roots 5,2,3,5,3,5,3,5,4,5,4,5,6,5,6,5,6,5,6,5,6,5,6,5,6; order 22,2,7,20,5,18,3,16,11,14,9,12,23,10,21,8,19,6,17,4,15,1,13

173: roots 2,3,5,3,5,4,6,5,6,5,6; order 1,4,10,2,8,6,11,5,9,3,7

176: roots 2,3,5,4,3,5,4,6,5,6,5,6,7,6; order 1,5,11,2,9,7,14,4,12,6,10,3,8,13

177: roots 5,2,3,5,3,5,3,5,4,5,4,6,5,6,5,6,5,6,5,6,5,6; order 21,2,7,19,5,17,3,15,11,13,9,22,10,20,8,18,6,16,4,14,1,12

179: roots 5,3,4,3,4,5,4,6,4,6,5,6,5,6,7,6,7; order 13,4,9,2,7,11,5,16,3,14,6,12,1,10,17,8,15

180: roots 3,5,3,5,3,5,3,5,4,6,5,6,5,6,5,6,5,6; order 7,17,5,15,3,13,1,11,9,18,8,16,6,14,4,12,2,10

181: roots 5,2,5,4,5,4,6,5,6,5,6,5,6; order 12,2,10,6,8,4,13,5,11,3,9,1,7

184: roots 2,4,6,4,6,7,6; order 1,4,7,2,5,6,3

185: roots 5,3,4,6,4,6,5,6,7; order 2,5,7,3,8,1,6,9,4

188: roots 5,4,2,4,6,4,6,5,6,7,6,7; order 3,6,8,4,11,2,9,1,7,12,5,10

194: roots 2,3,5,3,5,6,5,6,5,6; order 1,4,9,2,7,10,5,8,3,6

197: roots 5,3,6,4,6,5,6,7,6; order 4,2,9,6,7,1,5,8,3

198: roots 4,2,5,4,6,5,4,5,6,5,6,7,6,7; order 7,2,10,4,13,8,1,6,11,3,9,14,5,12

199: roots 4,2,4,6,4,6,4,5,4,5,6,7,6,7,6; order 9,2,7,15,5,13,3,10,1,8,11,14,6,12,4

200: roots 3,4,6,5,6,7,6,7; order 1,2,7,4,5,8,3,6

201: roots 3,4,3,4,6,5,4,6,7,6,7,6; order 7,3,12,1,4,10,2,6,8,11,5,9

202: roots 4,6,4,6,4,6,7,6,7,6; order 5,10,3,8,1,6,9,4,7,2

203: roots 4,6,3,4,6,4,6,7,6,7,6; order 3,6,11,4,9,1,7,10,5,8,2

204: roots 4,6,3,4,6,5,6,7,6,7,6; order 3,11,4,9,1,6,7,10,5,8,2

205: roots 4,2,3,4,6,5,4,5,6,7,6,7; order 7,2,3,4,11,8,1,6,9,12,5,10

206: roots 2,3,4,6,4,6,5,4,5,6,7,6,7,6,7; order 1,8,2,5,14,3,9,12,7,10,15,6,13,4,11

233: roots 2,6,4,5,6,5,6,5,6,7,6,7,6,7,6,7; order 1,15,8,13,3,6,11,4,9,16,7,14,5,12,2,10

234: roots 6,5,6,5,6,5,7,6,7,6,7,6,7; order 12,6,10,4,8,2,13,5,11,3,9,1,7

235: roots 2,6,3,5,6,5,6,5,7,6,7,6,7,6,7; order 1,14,3,8,12,6,10,4,15,7,13,5,11,2,9

236: roots 2,6,5,6,4,5,6,5,7,6,7,6,7,6,7; order 1,14,5,8,12,6,10,3,15,7,13,4,11,2,9

243: roots 4,5,4,6,5,7,6,7; order 3,7,5,1,2,8,4,6

245: roots 4,5,6,5,7,6,7; order 6,4,1,2,7,3,5

246: roots 6,3,6,5,4,5,6,4,6,5,7,6,7,6,7,6,7,6,7; order 8,2,18,5,16,10,14,6,12,4,19,9,17,7,15,3,13,1,11

247: roots 6,2,3,2,6,3,6,5,4,5,4,5,6,4,6,5,7,6,7,6,7,6,7,6,7; order 14,4,6,2,11,3,26,9,24,16,22,12,20,10,18,8,27,15,25,13,23,7,21,5,19,1,17

248: roots 6,2,6,5,4,5,4,6,4,6,5,7,6,7,6,7,6,7,6,7; order 9,2,7,19,5,17,11,15,6,13,4,20,10,18,8,16,3,14,1,12

249: roots 3,2,6,5,4,6,4,6,5,7,6,7,6,7,6,7; order 7,2,5,15,1,13,9,11,4,16,8,14,6,12,3,10

250: roots 2,6,5,4,6,5,7,6,7,6,7; order 1,4,10,6,8,3,11,5,9,2,7

251: roots 2,3,6,5,6,5,7,6,7,6,7; order 1,2,10,6,8,4,11,5,9,3,7

252: roots 2,6,4,6,5,6,5,6,5,7,6,7,6,7,6,7,6,7; order 1,17,9,15,3,13,7,11,5,18,8,16,6,14,4,12,2,10

253: roots 4,3,6,4,6,4,5,6,5,6,5,7,6,7,6,7,6,7,6,7; order 19,2,17,6,11,4,9,15,1,13,7,20,10,18,8,16,5,14,3,12

339: roots 3,6,4,3,6,8,7,8,7; order 4,9,1,7,3,5,8,2,6

341: roots 3,7,6,4,6,7,8,7; order 4,8,1,5,6,3,7,2

343: roots 5,4,6,4,6,5,6,8,7,8; order 9,4,6,2,7,1,5,10,3,8

344: roots 7,6,5,6,7,8,7,6,7,8,7,8; order 11,8,3,4,9,12,7,2,5,10,1,6

346: roots 5,7,4,6,5,6,8,7,6,7,8; order 3,9,5,6,1,10,4,8,11,2,7

347: roots 3,4,6,4,6,7,8,7,8; order 8,4,1,2,5,9,3,6,7

348: roots 6,5,6,5,6,7,6,7,8,7,8; order 10,4,7,2,5,11,3,8,1,6,9

349: roots 7,4,6,5,6,8,6,7,6,8; order 9,2,7,4,5,8,3,10,1,6

350: roots 7,4,3,4,6,8,6,8; order 3,4,7,2,5,8,1,6

353: roots 3,2,6,3,5,4,6,7,6,8,7,8; order 2,4,6,1,11,9,5,7,12,3,8,10

354: roots 7,4,3,4,6,8,7,6,8,7,8; order 10,3,4,8,2,5,11,7,9,1,6

358: roots 3,2,7,4,5,6,8,6,7,8; order 2,4,1,8,5,6,9,10,3,7

363: roots 7,6,4,6,5,6,7,8,7,8,7,8; order 11,3,5,6,12,9,4,7,2,10,1,8